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- ▶ We will stick with integer or finite state space

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- ▶ The row corresponds to present state and the column corresponds to next state.

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- ▶ In the ctmc counterpart for these examples, imagine the coin tosses itself/ dice rolls itself after waiting in the state for a random time that is exponentially distributed. (more later)

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- ▶ $i, j \in \mathcal{S}$ which is countable and $|\mathcal{S}| \leq \infty$

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- ▶ Y_n is geometric random variable.
- ▶ What would be the time spent in a state for a continuous time Markov chain ?